

Software Engineering

Exercise and Lab: Activity Diagram

1. Check-in System

When a passenger checks in, he or she first shows his or her ticket at the check-in counter. The ticket will be checked for its validity. If the ticket is not OK, the passenger will be referred to customer service. If the ticket is OK the passenger will check his or her luggage. If the luggage has excess weight, he or she will pay an additional fee. The luggage will be forwarded to baggage transportation. The passenger receives his or her boarding pass.

Develop an activity diagram to model the process of this system.

2. ATM

First, the customer inserts his or her ATM card. Second, the customer enters a five-digit PIN using the keypad. If the user enters the correct PIN, the screen displays the main menu. If the user enters an incorrect PIN, the screen displays an appropriate message, then the ATM restarts the authentication process (enter and check the PIN code). In the main menu there are withdraw, deposit and check balance option. If user select the withdraw money, ATM will display of menu of standard withdrawal amounts and an option to cancel the transaction. The customer enters a menu selection using the keypad and input the preferred withdrawal amount. If the withdrawal amount is greater than the customer's account balance, the screen displays a message stating this and telling the user to choose a smaller amount. If the withdrawal amount chosen is less than or equal to the customer's account balance (an acceptable amount), the ATM proceeds to next step.

If the customer chooses to cancel, the ATM displays the main menu and waits for customer input. The next step is the ATM return the card to the customer. When the customer take out the card, the ATM debits the withdrawal amount from the customer's account in the bank's database at the same time that the cash dispenser dispenses the desired amount of money to the customer; the screen displays a message reminding the customer to take the money.

Develop an activity diagram to model the process of this system.

3. Design the activity diagram in order to facilitate the process of making the lemon ginger cocktail:

Recipe:

First, you need to prepare all the ingredients (70g of ginger and 1 lemon). Then, you need to rush ginger and prepare the lemon juice from the lemon at the same

time. After that, mix that two together, and keep it for 30 minutes before adding the water. Lastly, you can filter it before drinking, or just drink it without filter.

4. Modeling the functionalities in a purchasing office by taking the following activities into account:

- Demand for the product
- Verifying the availability of that product (products are in stock)
- Processing the payment and preparing the receipt for customers
- Making products available to customers